Journal of Cleaner Production 582 (2026) 149786

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Root-associated microbial community dynamics in tobacco following topping: implications for nicotine degradation

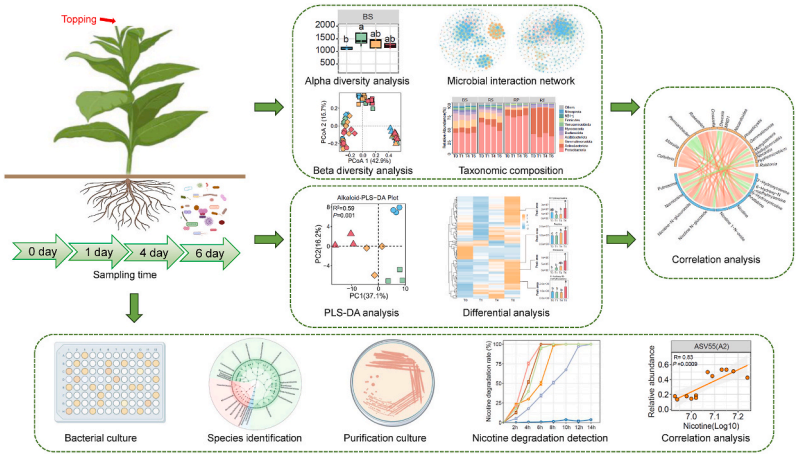
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HIGHLIGHTS

- Soil nicotine accumulation poses a significant environmental challenge.
- An increase in root nicotine content was observed following topping.
- Some bacteria and fungi were correlated to nicotine and nicotine derives.
- Five novel nicotine-degrading bacteria were isolated and validated.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Topping
Microbiome assembly
Alkaloids
Phytohormones
Nicotine-degrading microorganisms

ABSTRACT

Agricultural topping is followed by rapid nicotine synthesis in tobacco roots, which may pose environmental risks due to nicotine accumulation in soil. Microbial degradation offers a promising remediation strategy, but the temporal associations between nicotine accumulation and root-associated microbiomes remain unclear. Here, we integrated multi-omics analysis and experimental validation to characterize the post-topping temporal dynamics of tobacco root-associated microbiomes and their potential roles in nicotine degradation. Amplicon sequencing revealed distinct temporal shifts in bacterial communities in bulk soil and rhizosphere (explaining 2.84% of bacterial variance), as well as the fungal community in the root endosphere following topping. Notably, this

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https://doi.org/10.1016/j.jclepro.2026.149786
Received 22 July 2025; Received in revised form 20 October 2026; Accepted 21 October 2026
Available online 23 October 2026
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included enrichment of key nicotine-degrading microorganisms such as *Paenarthrobacter*, *Streptomyces*, *Arthrobacter*, *Ensifer*, and *Nocardioides*. Metabolomic profiling showed increased levels of nicotine, its biosynthetic precursor (putrescine), and microbial degradation products (6-hydroxynicotinine and 6-hydroxy-N-methylmyosmine) following topping. Integrated microbiome-metabolome analysis highlighted strong correlations between rhizosphere bacteria and nicotine-related alkaloids, particularly with nicotine-degrading genera such as *Paenarthrobacter*, *Streptomyces*, *Arthrobacter*, *Ensifer*, and *Nocardioides*. High-throughput cultivation yielded 221 and 197 nicotine-utilizing bacteria from rhizosphere and root, respectively. Functional validation of five representative strains demonstrated exceptional degradation efficiency, notably in *Ensifer* sp. A4 and *Pseudomonas* sp. B11. Given the absence of matched untopped controls in the field sampling scheme, our findings should be interpreted as temporal post-topping associations rather than causal evidence of topping effects. This study provides novel insights into the ecological patterns and temporal dynamics of tobacco root-associated microbiomes following topping, highlighting the potential role of specific microbes in nicotine degradation.

1. Introduction

Nicotine, the predominant alkaloid in *Nicotiana tabacum*, presents a significant environmental and public health challenge. Its inherent toxicity, addictive properties, and potential carcinogenicity pose well-documented risks to human health (Benowitz, 2010; Campain, 2004). Furthermore, the substantial quantities of nicotine-laden waste generated during tobacco processing can contaminate water sources and soils, leading to long-term detrimental effects on ecosystem function and biodiversity (Qiu et al., 2014; Tang et al., 2009). Consequently, the development of effective and sustainable strategies for nicotine degradation is of paramount importance, offering opportunities to mitigate environmental pollution and promote sustainable waste management practices.

At present, nicotine degradation strategies encompass physical adsorption (Lazarevic et al., 2011), photocatalytic degradation (de Franco et al., 2014) and microbial degradation (Wang et al., 2011). Among them, microbial degradation is increasingly recognized as a promising approach due to its high efficiency, environmental compatibility and cost-effectiveness (Wang et al., 2005). Nicotine-degrading microorganisms can utilize nicotine as a sole source of carbon and nitrogen, thereby effectively reducing nicotine concentrations in tobacco and contaminated environments (Gurusamy and Natarajan, 2013; Li et al., 2010). This dual potential, improving tobacco product safety and enabling eco-friendly waste treatment, has spurred significant interest in these microorganisms (Gurusamy and Natarajan, 2013; Li et al., 2010). To date, several nicotine-degrading bacteria, including *Arthrobacter, Paenarthrobacter, Streptomyces, Ensifer, Pseudomonas, Nocardioides, Sphingomonas* and *Rhodococcus*, and a limited number of fungi, such as *Aspergillus oryzae*, have been identified (Meng et al., 2010; Mu et al., 2020). However, the diversity of known nicotine-degrading microorganisms remains limited, and most reported strains were isolated from *ex situ* environments such as soil, activated sludge, or tobacco-processing waste. In contrast, the tobacco root-associated microbiomes, especially under conditions of elevated nicotine accumulation, remain largely unexplored as a source of novel nicotine degraders.

Nicotine is primarily synthesized in tobacco roots and is affected by developmental stages, phytohormones and environmental stresses (Sui et al., 2019). Topping, a common agricultural practice in tobacco cultivation (Gu et al., 2023; Li et al., 2007), has been shown to be associated with changes in alkaloid biosynthesis and phytohormone balance (Kajikawa et al., 2017; Shoji and Hashimoto, 2011; Tang et al., 2012). Several studies have demonstrated a substantial increase in nicotine content following topping (Sui et al., 2021; Zhang et al., 2023), accompanied by alterations in phytohormones such as jasmonic acid (JA), gibberellin (GA), auxin (indole-3-acetic acid, IAA), ethylene, and abscisic acid (ABA), all of which are known to regulate nicotine synthesis (De Boer et al., 2011; Hou et al., 2010; Lackman et al., 2011; Shoji et al., 2000). These physiological changes may in turn affect the root-associated microbial community, potentially favoring taxa with nicotine-metabolizing capacity. However, previous studies on topping have mainly focused on plant physiology and alkaloid regulation, while its effects on root microbiome assembly and function remain poorly understood. In particular, whether nicotine-enriched roots after topping can recruit putative nicotine-degrading microorganisms has not been investigated.

Plant root-associated microbiomes play a crucial role in plant adaptation to environmental stresses (Li et al., 2024; Zavaliev and Dong, 2024). Plants can actively modulate their microbial taxa by altering root exudate composition (Hong et al., 2022), thereby recruiting specific microbes under nutrient deficiency or stress conditions (Liu et al., 2024; Santoyo, 2022; Xuan et al., 2024). Therefore, we hypothesize that the substantial accumulation of nicotine in tobacco roots following topping may contribute to temporal changes in the root-associated microbiomes, possibly favoring taxa with nicotine-degrading potential. However, evidence linking post-topping nicotine and phytohormone dynamics to the enrichment of nicotine-metabolizing microorganisms is still lacking. Tobacco root samples, especially those collected after topping, may represent a promising but untapped niche for discovering novel nicotine-degrading strains.

Taken together, the tobacco root-associated microbiomes after topping have not been systematically examined as a reservoir of nicotine-degrading microorganisms, and the relationship between topping-induced changes in root nicotine and phytohormone profiles and shifts in microbial community structure remains unclear. To address these knowledge gaps, we investigated the dynamic changes in root-associated alkaloid and phytohormone profiles following topping, as well as the subsequent shifts in the root-associated microbiomes. The primary objectives were: (1) to elucidate the post-topping temporal dynamics of root-associated microbiomes; (2) to determine the potential associations between root alkaloid and phytohormone contents and root-associated microbiota; (3) to isolate and characterize novel microbial resources for nicotine degradation.

2. Materials and methods

2.1. Experiment design and sampling

Field experiments were conducted at the experimental station (35°12′N, 113°57′E) in Xinxiang, Henan Province, China, during the 2023 growing season (March to July). Tobacco (*Nicotiana tabacum* L. cv. Yunyan87) plants were cultivated under standard agronomic practices. Uniform plants (height: 116 ± 15 cm; leaf number: 15 ± 3) were selected and topped at 40–50% flowering (BBCH: 64–65) (Papenfus and Billenkamp, 2024), a well-defined physiological transition stage that strongly affects root metabolism, exudation, and rhizosphere microbial assembly. The topping procedure involved manual removal of the terminal inflorescence and the small, undeveloped leaves immediately below the inflorescence. To investigate the temporal dynamics of root-associated microbiomes following topping, samples were collected at multiple time points after the topping treatment.

Sampling was conducted at four defined time points: T0 (prior to topping), T1 (one day post-topping), T4 (four days post-topping), and T6

(six days post-topping), to capture immediate and early microbial responses to topping. Destructive sampling was required because repeated sampling from the same plant would unavoidably damage the root system and alter root exudation. For each biological replicate (n = 5), five plants were pooled into one composite sample using a systematic five-point method, reducing soil spatial heterogeneity and plant variability. Samples from the same ecological niches (BS, bulk soil; RS, rhizosphere; RP, root rhizoplane; RT, root interior) were collected from these five plants and pooled to form a single composite sample. This process was repeated to generate five independent biological replicates for each niche at each time point, resulting in a total of 80 samples (4 time-points × 4 niches × 5 replicates). To minimize environmental interference, all sampling events were conducted under stable weather conditions at approximately the same time of day, and soil moisture and pH were measured at each time point. The detailed sampling procedure was conducted as follows: After uprooting the tobacco plant, approximately 200 g of bulk soil was taken from the rhizospheric zone (0–15 cm depth and 20 cm radial distance from the root-stem junction) using a sterilized stainless-steel auger (Tao et al., 2022). The loose soil and rocks adhering to the root surface were gently shaken off. Soil tightly adhering to roots (diameter < 3 mm) was obtained as the rhizosphere samples by carefully brushing with sterile soft-bristle brushes (Zhang et al., 2017). The root samples were transferred into 10 mL sterile phosphate-buffered saline (PBS) solution. After vortexing at 2000 rpm for 10 min and sonicating at 40 kHz for 2 min, the washing solution was taken and centrifuged at 12,000 rpm for 15 min to obtain rhizoplane samples (Edwards et al., 2015). The remaining roots were soaked in 75% ethanol for 7 min, 1% sodium hypochlorite solution for 3 min, and again in 75% ethanol for 1 min, and then rinsed three times with sterilized distilled water. Sterility of the final rinse water was confirmed by plating 100 μL onto tryptone soy broth (TSB) medium and incubating at 30°C for 3 d. This surface-sterilization procedure removed epiphytic microorganisms, ensuring that the root interior samples represented the true endophytic community. After that, the roots were ground using liquid nitrogen, and the root endosphere samples were obtained. All samples were placed in a −80°C refrigerator until DNA extraction and microbial isolation.

2.2. The measurement of soil physicochemical properties

The gravimetric method was employed to measure the water content in the soil. Soil was mixed with deionized water at a mass-to-volume ratio of 1:2.5. After a 30 min equilibration period, a suspension was formed, and the potentiometric method was used to determine the pH (Cheng et al., 2024). Organic matter (OM) and organic carbon (OC) were quantified using the $K_2Cr_2O_7$-H_2SO_4 oxidation-reduction colorimetric method. To detect the content of total potassium (TK) and total phosphorus (TP), soil samples were digested with NaOH. Subsequently, TP was detected by molybdenum-antimony anti-spectrophotometry, while TK was measured using flame photometry. Total nitrogen (TN) analysis employed micro-Kjeldahl digestion coupled with steam distillation. Nitrate nitrogen (NO_3^--N) was measured through steam distillation, and ammonium nitrogen (NH_4^+-N) was determined by indophenol-blue colorimetry. Available phosphorus (AP) was extracted using $NaHCO_3$ and subsequently measured by molybdate-blue colorimetry. Available potassium (AK) was measured using ammonium acetate extraction followed by atomic absorption spectrometry analysis, with ascorbic acid serving as the reductant (Wang et al., 2022). Exchangeable potassium (K), exchangeable sodium (Na), exchangeable calcium (Ca), and exchangeable magnesium (Mg) were extracted with 1 M ammonium acetate (pH 7.0) and determined by atomic absorption spectrometry (for Ca and Mg) and flame photometry (for K and Na). Exchangeable chloride (Cl) was measured by silver nitrate titration after water extraction (Bao, 2000).

2.3. DNA extraction and amplification sequencing

The DNA Extraction Kit (MO BIO, San Diego, CA, USA) was used to extract DNA from soil and root samples. The NanoDrop® ND-1000 spectrophotometer (NanoDrop Technologies, Wilmington, USA) was used to accurately measure the concentration and quality of the extracted DNA. For bacterial community analysis, the V5–V7 region of the 16S rRNA gene was amplified with universal primers 799F and 1193R (Table S1). Fungal communities were targeted by amplifying the ITS1 region with primers ITS1F and ITS2. After amplification, PCR products were purified to remove residual primers and enzymes. Final library DNA concentrations were quantified with Qubit 4.0 fluorometer (Thermo Fisher Scientific). Then, fragment size distribution and library integrity were verified via capillary electrophoresis on the Agilent Bioanalyzer 2100 platform. Finally, the high-quality library was sequenced by Illumina NovaSeq platform and the paired-end reads of 250 bp were obtained.

2.4. Amplification data analysis

Sequencing data for each sample were separated according to the barcode sequence of the sample and the PCR amplification primer sequence. FLASH (v1.2.11) was used to merge the raw paired-end reads (Magoc and Salzberg, 2011). Subsequently, Cutadapt software was employed to accurately match the reverse primer sequence and excise the remaining sequence from the amplicon sequencing data. Fastp (v0.23.1) was utilized to filter the tags (Bokulich et al., 2013), and chimeric sequences were eliminated by comparison against a reference database (Edgar et al., 2011). The DADA2 (Callahan et al., 2016) module in QIIME2 (v2023.2) (Estaki et al., 2020) was used to denoise the effective sequences and infer unique Amplicon Sequence Variants (ASVs). These ASVs were then taxonomically classified using the SILVA (v138) (Quast et al., 2013) database for bacteria and the UNITE (v2021.5.10) (Nilsson et al., 2019) database for fungi. ASVs annotated as mitochondria or chloroplasts were removed. Additionally, ASVs with a total abundance of less than 10 reads were removed from all samples. To prevent potential biases, ASVs covered by less than 2 sequences were also filtered.

2.5. Metabolites detection and analysis

For metabolomic analysis, three root biological replicates were randomly selected from the five available replicates per time point to ensure a representative subset. Ultra Performance Liquid Chromatography (UPLC) (ExionLC™ AD) combined with Tandem Mass Spectrometry (MS/MS) (QTRAP® 6500+) was employed to detect the contents of phytohormones (Pan et al., 2010) and alkaloids (Chen et al., 2013) in these tobacco roots, as well as soil nicotine content. Compound identification was achieved through spectral library matching against authenticated reference standards. Quantitative measurements employed Multiple Reaction Monitoring (MRM) mode with signal acquisition controlled at 2.5 ms dwell time per transition. Raw MS/MS data were processed through MultiQuant 3.0.3 software for peak integration and analyte quantification. Chromatographic peaks corresponding to the tested substances in different samples were integrated and corrected. Total ion chromatograms (TIC) were used to assess the overall quality of the chromatographic separation and the stability of the analytical system. The coefficients of variation (CV) were calculated to evaluate the reproducibility of the measurements. To further explore the differences in metabolite profiles between different samples, Partial Least Squares Discriminant Analysis (PLS-DA) was performed in R using the ropls package.

2.6. Isolation and identification of nicotine-degrading microorganisms

The limiting dilution method (Zhang et al., 2021) was used to isolate

individual nicotine-degrading bacteria. The bacteria were dispersed into 96-well cell culture plates in 1/10 nicotine medium. The nicotine medium was prepared by adding 2 g of pure nicotine (filter-sterilized through a 0.22 μm filter membrane) to 1 L of sterile mineral salts medium (MSM), resulting in a final concentration of 0.0123 mol/L. The MSM (autoclaved at 121°C for 20 min) was composed of 13.3 g/L $K_2HPO_4 \bullet 3H_2O$, 4 g/L KH_2PO_4, 0.2 g/L $MgSO_4 \cdot 7H_2O$, and 1 mL/L trace element solution (0.05 g/L $CaCl_2 \bullet 2H_2O$, 0.05 g/L $CuCl_2 \bullet 2H_2O$, 0.008 g/L $MnSO_4 \bullet H_2O$, 0.004 g/L $FeSO_4 \bullet 7H_2O$, 0.1 g/L $ZnSO_4$, and 0.1 g/L $Na_2MoO_4 \bullet 2H_2O$). After culturing for 16 days at room temperature, the hypervariable V5–V7 region of 16S rRNA genes of the bacteria was amplified using domain-specific primers (Table S1). Each primer was labeled with a unique barcode, which facilitated the identification of different bacterial strains during subsequent analysis. By comparing the obtained 16S rRNA gene sequences with known sequences in databases, the taxonomic information for each pure culture was obtained.

Then, we selected the strains with the highest tolerance and degradation efficiency based on color change and growth status for subsequent verification experiments. The DNA of nicotine-degrading bacteria was obtained using kit (Omega Biotek Inc., Doraville, GA, USA). The full-length 16S rRNA gene fragment was amplified using the universal primers 27F and 1492R (Table S1), followed by Sanger sequencing. MEGA X was used to construct phylogenetic trees via the Neighbor-Joining method (Kumar et al., 2018). These bacteria were streaked on TSB plates to observe their morphology. To determine the growth characteristics of nicotine-degrading bacteria, including catalase activity, cellulase activity, hemicellulase activity, and their ability to produce hydrogen sulfide (H_2S), melanin, siderophores, and phosphorus dissolution, we referred to the method of (Zhang et al., 2024). Nicotine-degrading bacteria sequences were compared with amplicon data by BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi).

Subsequently, these bacteria were inoculated into TSB liquid medium and incubated at 30°C. The growth of the bacteria was monitored by measuring the optical density at 600 nm (OD_{600}). The bacterial cultures were centrifuged at 8000 rpm for 10 min at 4°C when the OD_{600} reached 2.0. The resulting cell pellets were then washed twice with PBS to remove any residual medium components. After washing, the cells were resuspended in a 100 mg/L (0.6164 mmol/L) nicotine medium and incubated for 24 h to adapt the bacteria to the nicotine-containing environment. After the 24 h adaptation period, the bacterial strains were inoculated into a nicotine medium containing 2 g/L (0.0123 mol/L) of nicotine with a 5% (v/v) inoculum size. The 2 g/L nicotine concentration was used to apply sufficient selective pressure to isolate efficient degraders while remaining below the acute toxicity threshold. A sterile medium containing 2 g/L (0.0123 mol/L) of nicotine without bacterial inoculation was used as a blank control, which can distinguish biological degradation from abiotic nicotine loss. The cultures were then incubated at 30°C, and samples were taken at 2 h intervals for further analysis. At each sampling time point, the samples were centrifuged at 12,000 rpm for 10 min at 4°C. The supernatant was collected and used to determine the nicotine content by detecting the absorbance at 259 nm. The cell pellets after centrifugation were washed twice with sterile 1×PBS buffer, resuspended in an equal volume of PBS buffer, and OD_{600} was measured to determine the cell density.

After crushing the tobacco leaves, 1.0 g of the tobacco leaf powder was weighed and placed in a 50 mL conical flask. In the experimental group, 2 mL of the acclimated bacterial suspension ($OD_{600} = 1.0$) was added, while in the control group, the same volume was replaced with sterile PBS buffer. The mixture was then incubated in a 37°C incubator for 5 days. After the cultivation was completed, the samples were dried at 65°C, and then nicotine was extracted by the acidic extraction method. The specific extraction operation is as follows: 0.05 g of the dried sample was weighed, 0.25 g of activated carbon and 25 mL of 0.5 mol/L HCl solution were added, and the mixture was vortexed and shaken for 5 min, then filtered. The filtrate was diluted to 100 mL. The blank control was diluted to the same volume with 25 mL of HCl solution of the same concentration. The absorbance values at 236 nm, 259 nm, and 282 nm (A236, A259, and A282) were determined using a UV-Vis spectrophotometer. Nicotine content was calculated according to the formula: Nicotine content $= 1.059 \times [A259 - 0.5(A236 + A282)] \times 100 \times 100/[W \times (1 - H_2O\%) \times 34.3 \times 1000]$.

(1.059 is the correction factor, W is the dry weight of tobacco leaves, and 34.3 is the absorbance coefficient of nicotine standard solution).

2.7. Experimental design and validation for nicotine biodegradation

To strengthen the evidence of causality in the relationship between metabolomics and microbial communities, we designed and conducted a laboratory experiment using field soil. This experiment tested if exogenous nicotine alone could reproduce the observed microbiome changes after topping, thereby supporting nicotine as a root-secreted mediator of microbial assembly. Field soil with an initial moisture content of 30% was divided into three treatment groups: (1) control group (CK) receiving no nicotine, (2) treatment group XN20 supplemented with 20 mg/kg nicotine solution, and (3) treatment group XN30 supplemented with 30 mg/kg nicotine solution. Each treatment group comprised three independent biological replicates. For each replicate, 100 g of soil was thoroughly homogenized with the respective nicotine solutions or an equivalent volume of sterile water (for CK) to ensure uniform distribution of additives. The soils were maintained at room temperature (approximately 25°C) under controlled conditions throughout the experiment. Soil samples were collected at days 0, 2, 5, 9, 14, and 20 after treatment, with 10 g of soil taken from each replicate at each time point. The collected samples were immediately frozen in liquid nitrogen and stored at −80°C for subsequent analysis of nicotine content and amplicon sequencing to assess changes in the microbial community.

In order to estimate the nicotine degradation capacity of isolated *Ensifer* sp. A4, the strain was applied to assess degradation kinetics from 50 mg/kg to <1 mg/kg under typical field soil conditions. Field soil with an initial moisture content of 30% was subjected to 30 min of high-pressure sterilization at 121°C to ensure sterility, and then cooled to room temperature (approximately 25°C). The experimental design included three treatment groups: (1) CK: sterilized soil (100 g) supplemented with 50 mg/kg nicotine and no microbial inoculation. (2) N1: sterilized soil (100 g) co-inoculated with 50 mg/kg nicotine and 5 mL *Ensifer* sp. A4 suspension ($OD_{600} = 0.5$). (3) N2: sterilized soil (100 g) co-inoculated with 50 mg/kg nicotine and 15 mL *Ensifer* sp. A4 suspension ($OD_{600} = 0.5$). Each treatment group included three independent biological replicates. Soil mixtures were homogenized thoroughly to ensure uniform distribution of nicotine and bacterial inoculum. The soils were maintained at room temperature (approximately 25°C) under controlled conditions throughout the experiment. Soil samples were collected at days 0, 2, 5, 9, 14, and 20 after treatment, with 10 g of soil taken from each replicate for nicotine content analysis. These data were used to develop a back-of-the-envelope model to estimate the number of days of *Ensifer* sp. A4 activity required for nicotine reduction in natural conditions.

2.8. Statistical analysis

To ensure the reliability and robustness of observed changes in soil parameters over time, multiple comparison tests were performed. For each soil variable, differences between time points were assessed using three post-hoc tests: Tukey's Honestly Significant Difference (HSD), Fisher's Least Significant Difference (LSD), and Duncan's multiple range test, following a one-way ANOVA. Only soil variables that exhibited statistically significant differences ($P < 0.05$) across all three multiple comparison methods were considered to have undergone robust changes. This conservative consensus strategy reduced the probability of reporting chance differences arising from multiple comparisons.

All the amplicon analyses were carried out in R (v4.2.3). For alpha

and beta diversity analyses, sequence counts were rarefied to the minimum sequencing depth to standardize unequal sequencing effort across samples. Alpha diversity was evaluated using Chao1 richness and the Shannon diversity index. Beta diversity patterns were assessed based on Bray-Curtis dissimilarity (Ramette, 2007) and visualized via Principal Coordinates Analysis (PCoA) using the ggplot2 package. To identify differentially abundant genera across different post-topping stages, ASV counts were first agglomerated to the genus level. DESeq2 was then applied to the raw, unrarefied genus-level count table. The model design was set as ~ condition, and pairwise contrasts were performed to compare each post-topping stage (T1, T4 and T6) against the pre-topping baseline (T0). Genera were identified as differentially abundant based on an adjusted *P*-value (FDR) < 0.05 and an absolute log2 fold change ≥ 1. The resulting list of genera was filtered according to these criteria to ensure statistical robustness (Love et al., 2014).

For network construction, ASVs detected in fewer than three samples were removed to minimize noise. Spearman correlation analysis was performed among the ASVs. Statistically significant correlations ($|r| > 0.9$, and false discovery rate (FDR)-adjusted $P < 0.05$ using the Benjamini-Hochberg (BH) method) were retained, ensuring that the network edges represented strong and statistically reliable associations. Microbial co-occurrence networks were built with Gephi 0.9.7 (Bastian et al., 2009), where nodes corresponded to individual ASVs and edges represented significant pairwise correlations. Calculation of within-module (*Zi*) and among-module (*Pi*) connectivity allowed node categorization into peripherals ($Zi \leq 2.5$, $Pi \leq 0.62$), connectors ($Zi \leq 2.5$, $Pi > 0.62$), module hubs ($Zi > 2.5$, $Pi \leq 0.62$), or network hubs ($Zi > 2.5$, $Pi > 0.62$).

The ComplexHeatmap package was used to generate heatmaps visualizing metabolite differences across samples. Spearman's rank correlation analysis was used to assess the associations between the relative abundance of microbial taxa and the concentrations of nicotine-related alkaloids. To visualize significant correlations between microbial taxa and nicotine-related alkaloids, a chord diagram was generated. To ensure statistical robustness and minimize the risk of false positives, only correlations with an absolute Spearman correlation coefficient $|r| \geq 0.6$ and an adjusted *P*-value (FDR) < 0.05 (corrected via the Benjamini-Hochberg method) were retained for analysis. These significant correlations were visualized using a chord diagram generated by the circlize package. Pearson correlation coefficients and *P* values between log10-transformed nicotine content and the relative abundance of nicotine-degrading bacterial ASVs were calculated using the rcorr function from the Hmisc package. The correlation analysis included 12 paired samples per ecological niche (3 biological replicates × 4 time points) for which both metabolomic and microbiome data were available. Scatter plots with linear regression lines and 95% confidence intervals were generated using ggplot2. *P* values < 0.05 were considered statistically significant.

3. Results

3.1. Changes of soil physicochemical properties following topping

Principal component analysis (PCA) for soil physicochemical parameters showed that samples at different time points could be distinguished effectively, with the first two components explaining 39.78% and 18.58% of the variance, respectively (Fig. S1, Table S2). We validated the reliability of observed changes by requiring consistency across three multiple comparison tests (Tukey HSD, LSD, and Duncan; see Metho. Specifically, TN, AP, AK, K, Na, Cl, and WC showed statistically significant variations over the sampling period ($P < 0.05$, across all three tests; Table S2). In contrast, TP, TK, AN, ASK, SK, OC, OM, NH_4^+-N, NO_3^--N, Ca, Mg, and pH did not demonstrate consistently significant differences across all three methods, suggesting they remained relatively stable. However, we emphasize that the rapid fluctuations in several measured soil nutrient variables (e.g., TN, AP, AK) were notable and should be interpreted with caution, as they may reflect transient, localized perturbations rather than stable shifts.

3.2. Community assembly of bacterial and fungal microbiomes following topping

A total of 2,075,534 bacterial and 4,739,871 fungal high-quality reads were obtained from 80 samples. These reads were classified into 5837 bacterial and 1188 fungal ASVs, respectively. The results of alpha diversity indexes showed that the alpha diversity of bacterial communities was significantly decreased from bulk soil to root endosphere in terms of Chao1 richness index and Shannon diversity index ($P < 0.05$). For fungal communities, only the alpha diversity in the root endosphere was significantly lower than that in the other three compartments ($P < 0.05$) (Fig. S2A). In bulk soil, the bacterial Chao1 richness index significantly increased following topping, followed by a gradual decline, whereas the fungal Chao1 richness index in both bulk soil and the root endosphere continued to increase following topping ($P < 0.05$) (Fig. 1A, Tables S3 and S4). The Shannon diversity index of bacterial community in rhizosphere and fungal community in bulk soil increased significantly following topping. However, the Shannon diversity index of bacterial community in the rhizoplane showed a trend of first increasing and then decreasing (Fig. S2B, Tables S3 and S4).

PCoA was used to assess differences in microbiome composition among different samples. The first two components explained about 59% of the variation in bacterial communities and about 44% of the variation in fungal communities. Permutational multivariate analysis of variance (PERMANOVA) calculated the relative contributions of niche and temporal dynamics to root-associated microbiota structure. The results showed that niches accounted for a larger proportion of the variation in microbial community structure (bacteria: 43.77%, fungi: 24.87%, $P < 0.001$), while temporal changes following topping were associated with a relatively smaller proportion of the observed variation in root-associated microbiomes (bacteria: 2.84%, fungi: 2.54%, $P < 0.001$) (Fig. 1B). PERMANOVA and PCoA of each niche microbiome showed that there were significant temporal shifts in the microbial communities in different niches (Fig. 1C). The bacterial and fungal communities in the soil (bulk soil, rhizosphere soil, and rhizoplane soil) changed significantly following topping (PERMANOVA; $P < 0.05$). In the root endosphere, only fungal communities changed significantly following topping (PERMANOVA; $P < 0.05$) (Fig. 1C).

3.3. Temporal dynamics of microbial interkingdom co-occurrence networks following topping

To assess the temporal variations in microbial associations, we performed bacterial-fungal co-occurrence network analysis across different time points following topping. Our findings indicated that microbial interkingdom network patterns exhibited dynamic shifts over the observation period (Fig. 2). Specifically, the number of nodes and edges varied across time points (T0: 254 nodes and 1003 edges, T1: 247 nodes and 1118 edges, T4: 197 nodes and 676 edges, T6: 207 nodes and 835 edges) (Fig. 2A). Compared with T0 (7.08%), the proportion of negative co-occurrence edges increased following topping (T1: 18.43%, T4: 13.31%, T6: 17.6%), which may statistically indicate a potential increase in complex microbial associations. Following topping, the number of bacterial nodes in the network first increased and then decreased, while the number of fungal nodes showed a declining trend (Fig. 2A). Moreover, the network connectivity of bacterial taxa increased first and then decreased over time, while fungal taxa exhibited the opposite trend (Fig. 2B). In addition, bacteria-bacteria (BB) interactions were more frequent after the initial time point, while bacteria-fungi (BF) and fungi-fungi (FF) interactions were fewer (Fig. 2C). These findings suggest that bacterial communities exhibited more pronounced temporal variations in their association patterns than fungal communities following topping.

Topological stratification based on *Zi* and *Pi* connectivity

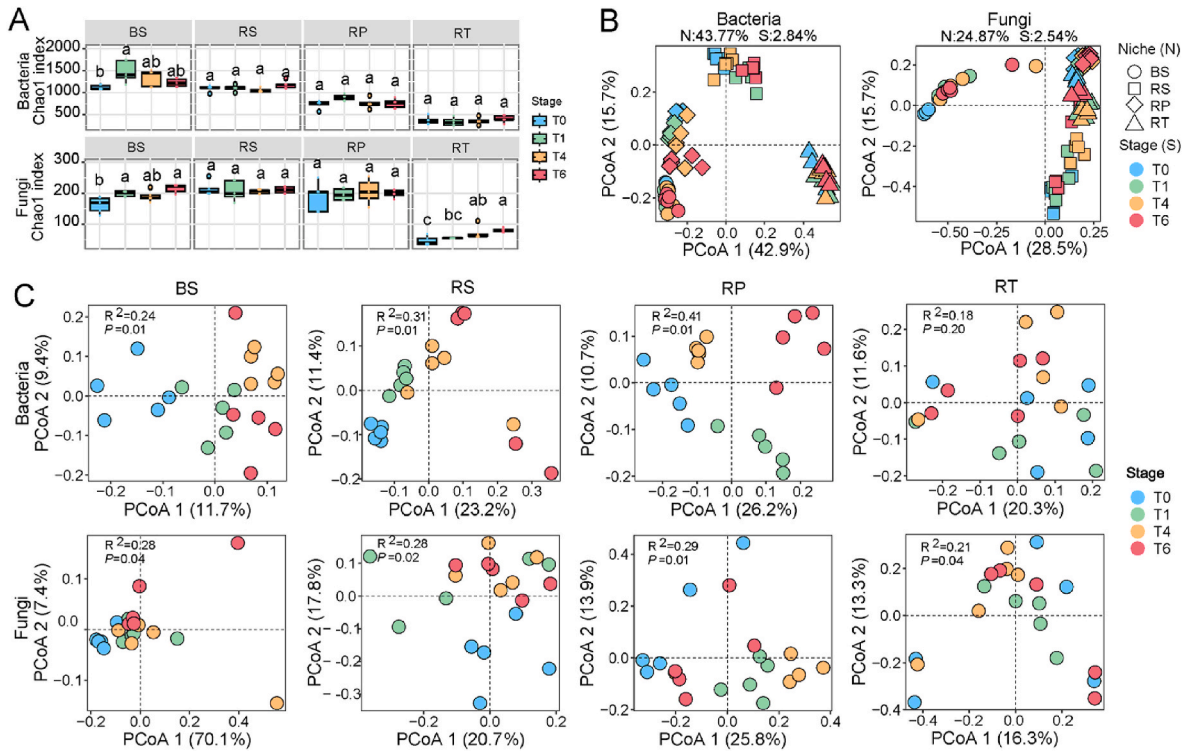


Fig. 1. Alpha and beta diversity of root-associated microbiomes in tobacco over the post-topping period. (A) Chao1 diversity index of bacterial and fungal communities in different niches of tobacco at different post-topping stages. Non-parametric Kruskal-Wallis test indicated statistically significant differences between different groups, denoting by different letters ($P < 0.05$). (B) PCoA was carried out using Bray-Curtis distance matrices to illustrate the distribution patterns of microbial communities across all samples. PERMANOVA was used to evaluate the relative contributions of niche (N) and stage (S) to the observed community differences. (C) PCoA was carried out using Bray-Curtis distance matrices to illustrate the distribution patterns of microbial communities in each niche.

demonstrated that the composition of putative key nodes in the co-occurrence network was different. Specifically, before topping (T0), the network contained 2 bacterial module hubs. During the post-topping observation period, the network composition was characterized by (T1: 2 bacterial module hubs and 1 fungal connector, T4: 3 bacterial module hubs and 1 fungal connector, T6: 2 bacterial module hubs and 5 bacterial connectors) (Fig. 2D–Table S5).

3.4. Temporal dynamics of microbiome composition following topping

Taxonomic classification showed slight changes in microbial communities at the phylum level following topping (Fig. 3A and B). The bacterial communities in each niche were mainly composed of *Proteobacteria*, *Actinobacteriota*, *Gemmatimonadota*, *Acidobacteriota*, *Bacteroidota*, *Myxococcota* and *Verrucomicrobiota* (Fig. 3A). The fungal communities in each niche were mainly composed of *Ascomycota*, *Basidiomycota*, *Mortierellomycota*, and *Glomeromycota* (Fig. 3B). The differences in the relative abundance of taxa observed after topping were determined by differential abundance analysis at the genus level. In the bulk soil, *Sphingomonas* exhibited a significant reduction following topping, while *Paenarthrobacter* demonstrated a proportional increase. The rhizosphere was characterized by an enrichment of *Streptomyces*, *Bacillus*, and *Arthrobacter*; the rhizoplane by an enrichment of *Ensifer*; and the root endosphere by a depletion of *Pseudomonas* and an enrichment of *Nocardioides* (Fig. 3C–Table S6). These bacteria, which showed enrichment or depletion following topping, were confirmed as nicotine-degrading bacteria in previous reports (Liu et al., 2015; Mu et al., 2020).

3.5. Changes in alkaloids and phytohormones biosynthesis in tobacco roots following topping

To investigate temporal dynamics of phytohormones and alkaloid in tobacco roots following topping, LC-MS was used to detect the content of phytohormones and alkaloids in tobacco roots. A total of 50 phytohormones and 236 alkaloids were detected in tobacco roots. PLS-DA showed that the composition of phytohormones (Fig. 4A) and alkaloids (Fig. 4B) differed across post-topping stages ($P < 0.05$). The first two principal components explained 44.9% and 53.3% of the variance in phytohormone and alkaloid composition, respectively. Compared with the tobacco plants before topping, the roots exhibited distinct accumulation patterns of phytohormones (Fig. S3) and alkaloids (Fig. 4C) in the post-topping phase. Among the phytohormones, the levels of ABA and IAA increased significantly on the fourth and sixth days following topping, while the content of GA8 decreased. In addition, the content of JA increased rapidly on the first day following topping, and then decreased (Fig. S3). Among the alkaloids, the root samples on the sixth day following topping showed the highest number of differentially accumulated alkaloids (up: 40, down: 7), followed by the first day (up: 5, down: 19) and the fourth day (up: 8, down: 7) (Fig. 4C). Heatmaps of the relative content of alkaloids at different periods of topping also showed that the content of most alkaloids increased on the sixth day following topping (Fig. 4D). The content of nicotine increased significantly on the fourth day following topping. The content of putrescine, a precursor of nicotine synthesis, increased significantly on the sixth day following topping. Additionally, the levels of 6-hydroxynicotine and 6-hydroxy-N-methylmyosmine—reported microbial transformation products of nicotine—also increased significantly on the sixth day following topping. (Fig. 4).

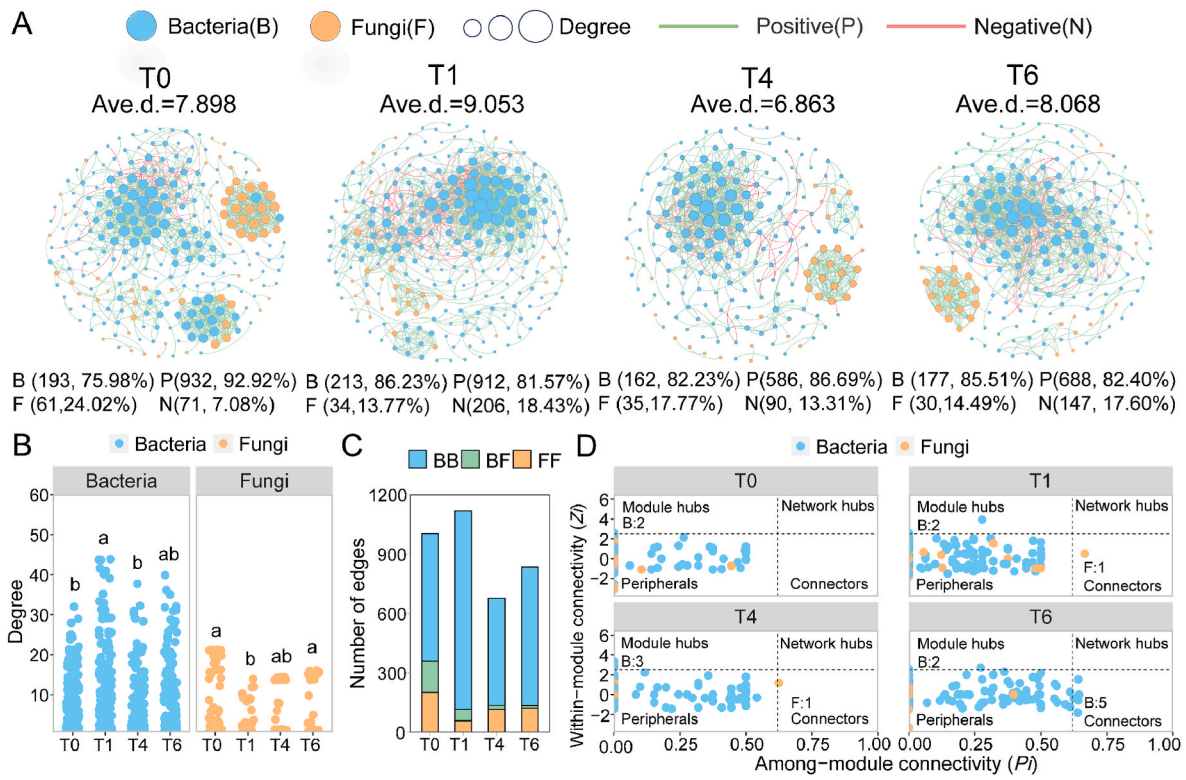


Fig. 2. Bacterial-fungal interkingdom co-occurrence networks. (A) Co-occurrence network analysis of full dataset showing microbial interkingdom network patterns at different stages of topping. "B" and "F" denoted bacteria and fungi, respectively. "P" and "N" denoted positive and negative correlations, respectively. (B) The node degree of bacteria and fungi at different stages of topping. Different letters indicated a significant difference determined by ANOVA test. (C) Number of bacterial–bacterial (BB), bacterial–fungal (BF), and fungal–fungal (FF) correlations in networks at different stages of topping. (D) The topological role of each ASV was determined according to the scatter plot of within-module connectivity (*Zi*) and among-module connectivity (*Pi*).

3.6. Correlations between microbial taxa and alkaloid contents in tobacco roots

We further used correlation analysis to study potential relationships between microbial taxa and nicotine-related alkaloids. The results showed that rhizosphere bacteria were more correlated with nicotine-related alkaloids compared to other niches. Nicotine, nicotine N-glucoside (a nicotine derivative), and putrescine (a nicotine-related precursor) showed more correlations than other alkaloids (Fig. 5, Table S7). In bulk soil, these compounds exhibited predominantly negative correlations (6, 5 and 5 negative *vs.* 3, 1 and 1 positive associations, respectively) with bacterial genera (Fig. 5A–E, F); notably, the nicotine-degrading bacterium *Paenarthrobacter* was positively correlated with nicotine and nicotine-N-glucuronide (Fig. 5A; $P < 0.05$). Conversely, rhizosphere bacteria demonstrated predominantly positive correlations (9, 13 and 11 positive *vs.* 3, 6, and 6 negative) (Fig. 5B–E, F), as particularly reflected by the significant associations of *Arthrobacter* with 6 alkaloids, while *Streptomyces* significantly correlated with 7 alkaloids (Fig. 5B; $P < 0.05$). In the rhizoplane, these compounds were positively correlated with 3, 6 and 4 bacterial genera respectively, and negatively with 2, 2 and 1 genera respectively (Fig. 5C–E, F). The nicotine-degrading bacterium *Ensifer* in the rhizoplane was significantly and positively correlated with putrescine (Fig. 5C; $P < 0.05$). In the root endosphere, nicotine, nicotine N-glucoside and putrescine primarily showed positive correlations with bacterial genera (Fig. 5D, E, F). The nicotine-degrading bacterium *Nocardioides* exhibited significantly positive correlations with 6 alkaloids, including nicotine (Fig. 5D; $P < 0.05$).

To further explore the associations between metabolomics and microbial communities, we conducted a soil microcosm experiment to examine changes in extractable nicotine concentrations and microbial communities following nicotine addition. Extractable nicotine concentrations were monitored over a 20-day period under different treatment conditions (Fig. 6A). Due to the absence of nicotine in the control group (CK), nicotine levels remained consistently low, with no apparent change in nicotine concentrations. In contrast, both nicotine-treated groups, XN20 (20 mg/kg nicotine) and XN30 (30 mg/kg nicotine), exhibited rapid and substantial decrease in extractable nicotine concentrations. Applied nicotine in the XN20 group was reduced to near-undetectable levels within 2 days and remained so until day 20. Similarly, the XN30 group showed a rapid decrease in nicotine content, reaching baseline levels by day 2 and maintaining them until day 20. It should be noted that the decrease in nicotine concentrations observed in the non-sterile field soil microcosms cannot be attributed solely to microbial degradation. The observed decline may also involve processes such as soil adsorption, volatilization, extraction-related effects, or abiotic transformation.

Concurrently, microbial community analysis revealed significant shifts following nicotine addition. *Paenarthrobacter* showed significant enrichment in both XN20 and XN30 treatments compared to CK from day 2 onwards ($P < 0.05$), with its relative abundance remaining significantly higher throughout the 20 days (Fig. 6B). *Arthrobacter* also displayed an enrichment trend in the nicotine-treated groups, with significantly higher relative abundance than CK from day 5 onwards ($P < 0.05$) (Fig. 6C). *Streptomyces* exhibited a significant increase in relative abundance in both XN20 and XN30 treatments starting from day 2 ($P < 0.05$) and maintained higher levels compared to CK throughout the experimental period (Fig. 6D). *Ensifer* also showed a notable increase in relative abundance from day 5 in the XN20 and XN30 treatments compared to CK ($P < 0.05$), with slightly higher abundance in XN30 at most time points (Fig. 6E). *Nocardioides* was significantly enriched mainly under the XN20 treatment condition ($P < 0.05$) (Fig. 6F).

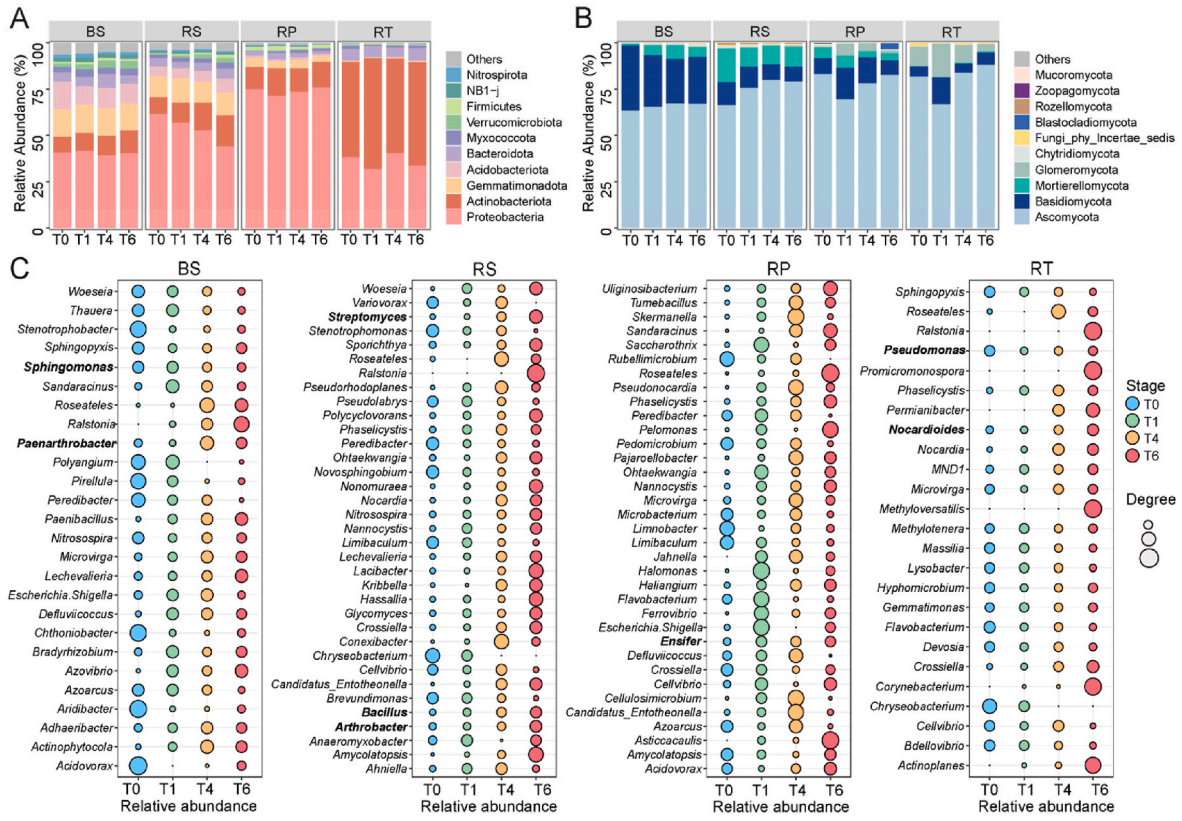


Fig. 3. Taxonomic composition and differential abundance analysis of the root-associated microbiomes at different post-topping stages. Relative abundance of bacterial (A) and fungal (B) communities at the phylum level. (C) Comparison of the top 35 genera which were significantly enriched or depleted following topping. Each circle represented a genus, and the size of each circle represented its relative abundance. Different colors represented different stages.

3.7. Identification of nicotine-degrading bacteria from topped samples

Given the substantial synthesis of nicotine in tobacco roots following topping, we investigated whether the abundance of nicotine-degrading microorganisms showed temporal changes following topping. Therefore, we attempted to isolate culturable bacteria capable of utilizing nicotine as the sole nitrogen and carbon source from topped samples. A total of 197 strains belonging to 55 genera were separated from the rhizoplane and root endosphere, and 221 strains belonging to 55 genera were isolated from the rhizosphere (Fig. 7A). In the rhizoplane and root endosphere samples, the most abundant genus was *Pseudomonas*, which made up 15.18% of the strains and was followed by *Trabulsiella*, *Achromobacter* and *Ensifer*, respectively. In contrast, in the rhizosphere samples, the most abundant genera were *Sphingobium* and *Sphingopyxis*, both accounted for 9.32% of the isolated strains. They were followed by *Arthrobacter*, *Ensifer* and *Mycobacterium* (Fig. S4).

Based on color changes and growth status, five strains exhibiting the best tolerance and degradation of nicotine were selected and identified as *Arthrobacter*, *Ensifer*, *Paenarthrobacter*, *Pseudomonas* and *Nocardioides* (Fig. 7B). These isolates exhibited distinct colony morphologies when grown on TSB agar medium (Fig. S5). The test for physiological and biochemical characteristics (Fig. S6) showed that the five nicotine-degrading strains exhibited catalase activity (H_2O_2 decomposition) and weak cellulase activity and could produce hydrogen sulfide (H_2S), while only *Arthrobacter* sp. A2, *Ensifer* sp. A4, and *Paenarthrobacter* sp. A6 showed hemicellulase activity. *Paenarthrobacter* sp. A6, *Pseudomonas* sp. B11 and *Nocardioides* sp. B15 could produce melanin. *Pseudomonas* sp. B11 and *Nocardioides* sp. B15 could produce siderophores, as indicated by color changes on CAS (Chrome Azurol S) agar. Besides, *Pseudomonas* sp. B11 and *Nocardioides* sp. B15 demonstrated phosphate solubilization on both organic and inorganic phosphorus media.

Analysis of ASV abundance revealed a significant post-topping increase in the relative abundance of these five strains (Fig. 7C, Fig. S7A), which coincided with the temporal accumulation of nicotine. Correlation analysis further revealed significantly positive correlations between nicotine content and the relative abundance of certain nicotine-degrading bacteria in root-related samples (Fig. 7D). Specifically, the relative abundance of *Arthrobacter* sp. A2 (ASV55) ($R^2 = 0.69$, $P = 0.001$) and *Ensifer* sp. A4 (ASV612) ($R^2 = 0.41$, $P = 0.027$) in the rhizoplane, and the relative abundance of *Pseudomonas* sp. B11 (ASV1156) ($R^2 = 0.74$, $P < 0.001$) and *Nocardioides* sp. B15 (ASV1264) ($R^2 = 0.59$, $P = 0.003$) in the rhizosphere, were significantly and positively correlated with nicotine content (Fig. 7D). Similar positive correlations were observed in the root endosphere for *Arthrobacter* sp. A2 (ASV55) ($R^2 = 0.46$, $P = 0.015$) and *Ensifer* sp. A4 (ASV612) ($R^2 = 0.79$, $P < 0.001$) (Fig. S7B).

The residual nicotine concentration over time in a liquid medium clearly demonstrated varying degradation efficiencies among the tested bacterial strains. Compared with CK, all tested bacterial strains, including *Arthrobacter* sp. A2, *Ensifer* sp. A4, *Paenarthrobacter* sp. A6, *Pseudomonas* sp. B11, and *Nocardioides* sp. B15, exhibited significant nicotine degradation (Fig. 8A). Nicotine degradation assays revealed that all five strains completely degraded 2 g/L of nicotine within 14 h (Fig. 8B). However, significant differences in degradation efficiency were observed. *Ensifer* sp. A4 and *Pseudomonas* sp. B11 exhibited the highest degradation rates, achieving complete nicotine depletion in 6 h (0.33 g/L/h), followed by *Nocardioides* sp. B15 (8 h, 0.25 g/L/h), *Arthrobacter* sp. A2 (10 h, 0.20 g/L/h), and *Paenarthrobacter* sp. A6 (14 h, 0.14 g/L/h) (Fig. 8B). Growth curves in nicotine-containing medium also revealed distinct growth patterns among the strains (Fig. 8C), with *Arthrobacter* sp. A2 and *Paenarthrobacter* sp. A6 exhibiting relatively rapid growth, while *Pseudomonas* sp. B11 exhibited relatively slow growth in nicotine-containing medium. To assess their potential for nicotine degradation in a more relevant context, the strains were applied

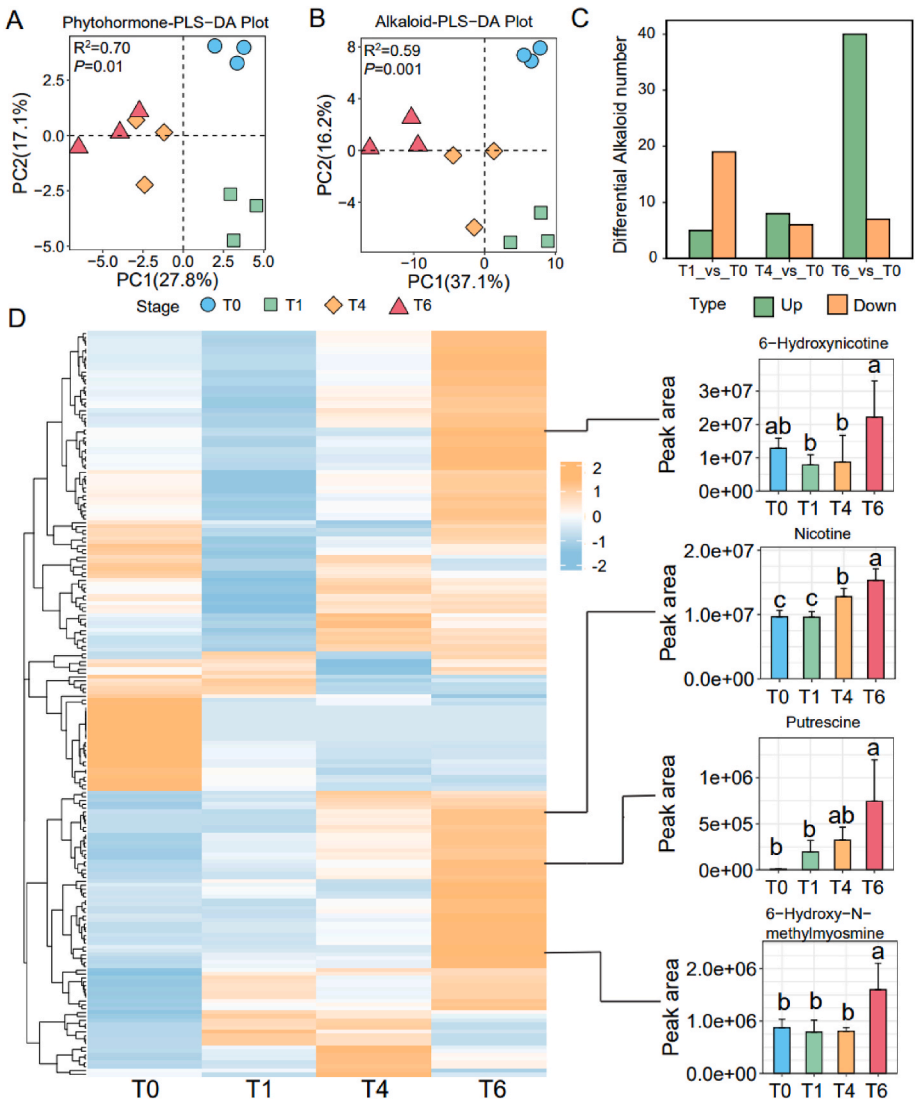


Fig. 4. Temporal dynamics of phytohormones and alkaloids in tobacco roots following topping. PLS-DA showing differences in phytohormones (A) and alkaloids (B) at different post-topping stages. (C) Differences in the abundance of alkaloids in tobacco roots at various time points compared to pre-topping levels. (D) Heatmaps showing the relative content of alkaloids in tobacco roots at different post-topping stages. The highlighted column charts showing the content of nicotine and nicotine related metabolites. Different letters indicated significant differences ($P < 0.05$) determined by the TukeyHSD test.

to tobacco leaves. After 5 days, all strains significantly reduced nicotine content in tobacco leaves ($P < 0.05$) (Fig. 8D). *Pseudomonas* sp. B11 exhibited the highest degradation rate (28.98%), followed by *Nocardioides* sp. B15 (26.30%), *Ensifer* sp. A4 (25.14%), *Arthrobacter* sp. A2 (19.38%), and *Paenarthrobacter* sp. A6 (18.81%), respectively (Fig. 8E). These results suggested the potential of these strains for bioremediation of nicotine in tobacco-related applications.

In addition, the degradation of nicotine in soil by the strain *Ensifer* sp. A4 was investigated under different inoculation levels. In the absence of *Ensifer* sp. A4 inoculation, nicotine exhibited slow degradation, indicating that its degradation under sterilized-soil microcosm conditions was minimal. In contrast, inoculation with either 5 mL (N1) or 15 mL (N2) of *Ensifer* sp. A4 significantly accelerated nicotine degradation compared to the control. A clear dose-dependent effect was observed, where higher inoculation levels of *Ensifer* sp. A4 led to faster and more complete nicotine removal from the soil (Fig. S8).

4. Discussion

Understanding temporal microbial changes following tobacco topping offers insights into the relationship between nicotine levels and the microbiome, which is essential for developing strategies to mitigate nicotine accumulation in soil. In this study, bacterial α diversity declined from bulk soil to root endosphere (Fig. S2A), consistent with the established root filtering models where root exudates selectively enriched specific commensal taxa through chemotactic and nutritional specialization (Edwards et al., 2015; Gu et al., 2024). Beyond this spatial filtering, topping also drove temporal shifts in microbial communities. Following topping, we observed temporal changes in α diversity (Fig. 1A, Fig. S2B), likely related to the subsequent release of novel root exudates associated with community reshaping (Badri et al., 2009). While PCoA and PERMANOVA analyses identified ecological niche as the primary factor shaping the microbial community (Fig. 1B), niche dynamics exhibited distinct temporal patterns following topping. This temporal reconfiguration, potentially linked to plant physiological responses to topping, distinguished our findings by showing how management practices created specialized microenvironments rather than merely observing static niche partitioning. These unique niches may provide preferential habitats for specific microbial taxa, including those potentially involved in nicotine degradation. The differential distribution patterns across these niches (Fig. 3C) underscored the complex co-occurrence patterns and the potential influence of plant management

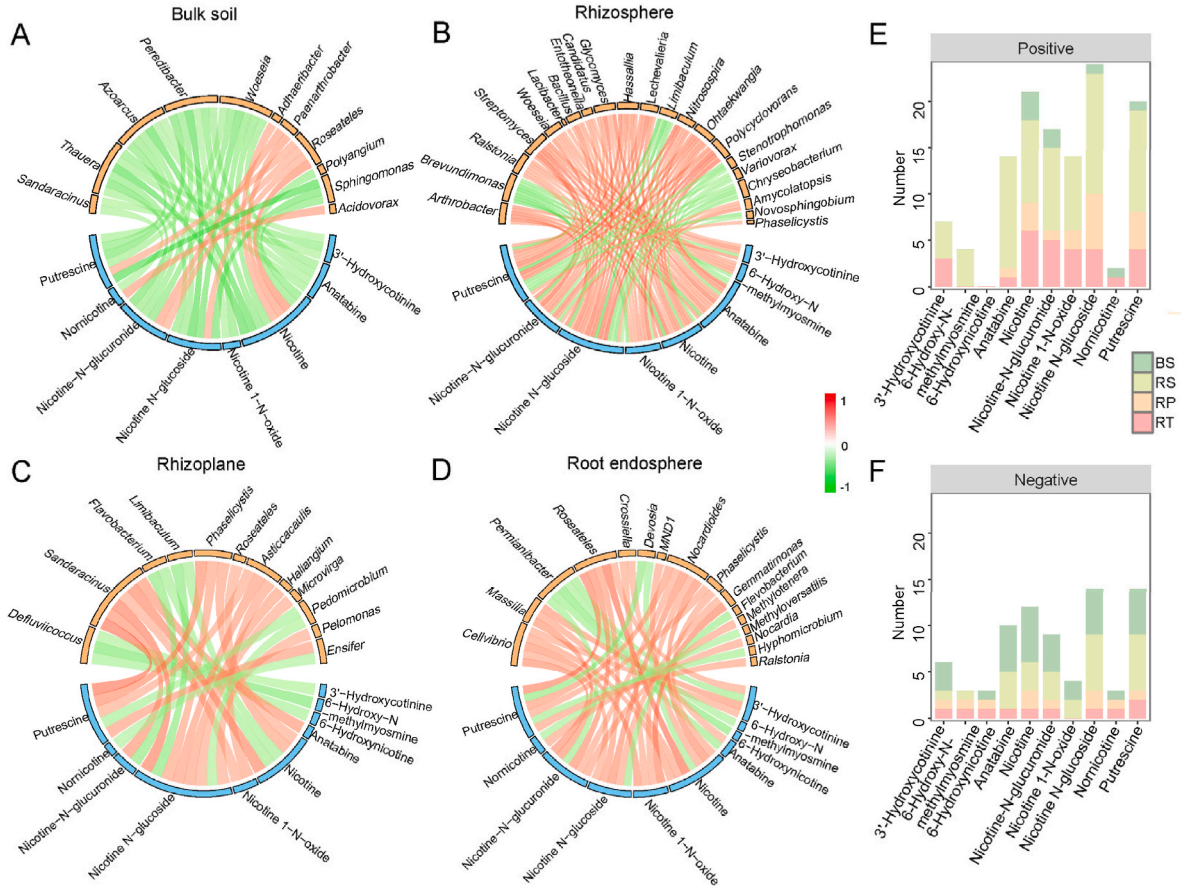


Fig. 5. Correlation between microorganisms and nicotine-related alkaloids. (A–D) Correlation analysis between microorganisms in different ecological niches and nicotine-related alkaloid content in tobacco roots. The red edges indicated positive correlations and blue edges indicated negative correlations. Only correlations with an absolute Spearman correlation coefficient $|r| \geq 0.6$ and an adjusted *P*-value (FDR) < 0.05 (calculated using the Benjamini-Hochberg method for multiple comparisons) were displayed in the figure. The number of bacterial genera that showed significantly positive (E) and negative (F) correlations with nicotine-related alkaloids across different niches.

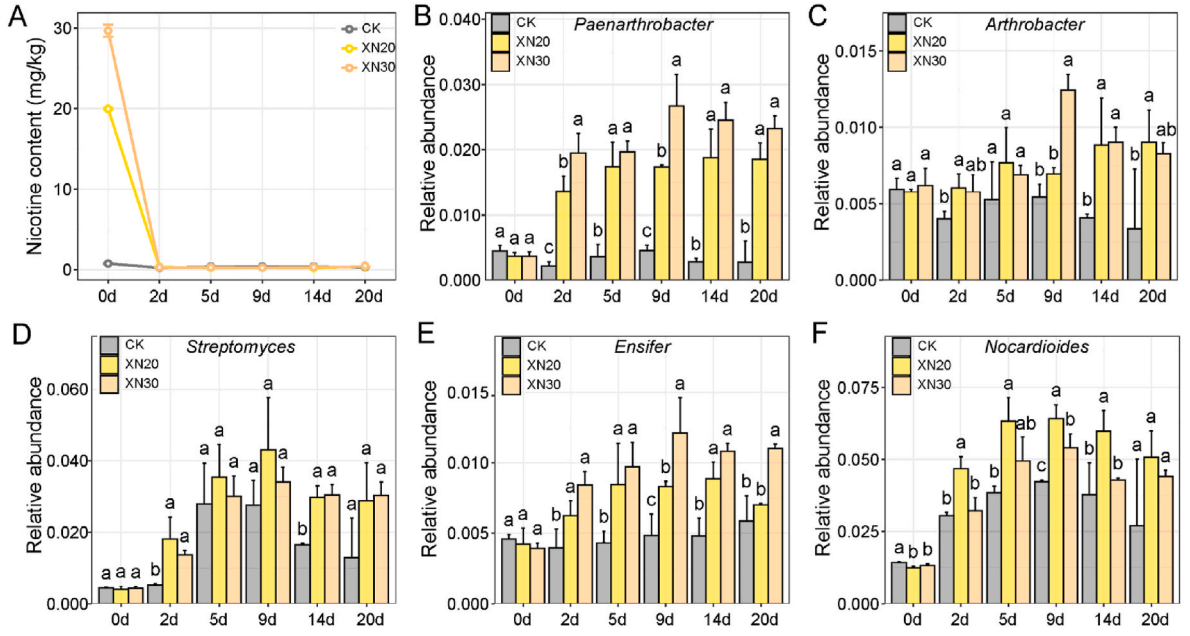


Fig. 6. Changes in extractable nicotine concentrations and enrichment of potential nicotine-degrading bacteria. (A) Decline curve of extractable nicotine in soil. (B–F) The enrichment patterns of nicotine-degrading bacteria under different nicotine concentrations. CK: soil with no nicotine. XN20: soil contained 20 mg/kg nicotine. XN30: soil contained 30 mg/kg nicotine.

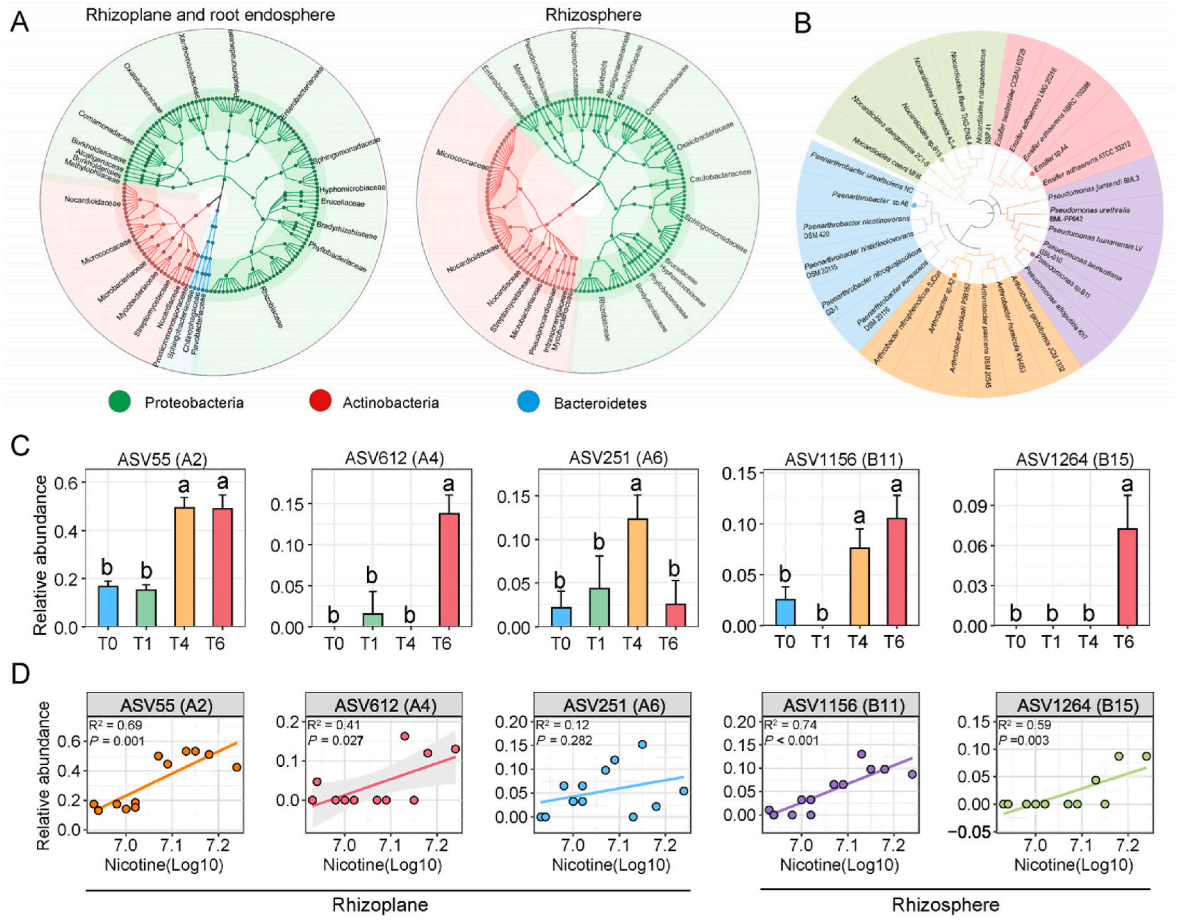


Fig. 7. Nicotine-degrading bacteria isolated from tobacco root-related samples. (A) Phylogenetic tree of culturable nicotine-degrading bacteria isolated from tobacco root-related samples collected after topping. (B) Phylogenetic analysis of nicotine-degrading bacteria based on full-length 16S rRNA sequences. (C) The relative abundance of five nicotine-degrading bacterial ASVs was significantly increased following topping. (D) Pearson's correlation analysis between the relative abundance of nicotine-degrading bacteria and nicotine content in tobacco roots. The analysis was performed using 12 paired samples per ecological niche (3 biological replicates × 4 time points) with matched metabolomic and microbiome data.

practices on microbial community assembly. In essence, while niche partitioning was fundamental to microbial assembly (Bulgarelli et al., 2013), the post-topping temporal shifts in the microbial communities were closely linked to these niche dynamics, potentially facilitating the presence of some functional microbes. Understanding these integrated plant-microbe-metabolite interactions is paramount for developing targeted strategies to mitigate nicotine accumulation and promote soil health in tobacco cultivation.

The observed temporal changes in the microbial co-occurrence networks following topping suggested dynamic shifts in co-occurrence patterns (Fig. 2). The increase in negative correlations following topping may indicate potential shifts in resource partitioning or niche segregation among microorganisms, possibly related to altered resource availability or antimicrobial compounds produced by the plant or other microorganisms (Foster and Bell, 2012). While competition can be a prevalent interaction among microbes (Foster and Bell, 2012), our correlation-based analysis does not directly confirm such ecological interactions. Regarding network topology, the more pronounced alterations in bacterial node numbers and connectivity compared to fungi may reflect the diverse metabolic capabilities and faster generation times of bacterial taxa. The observed increase in bacteria-bacteria co-occurrence links following topping may indicate potential shifts in resource utilization or the emergence of cooperative consortia after environmental disturbances (Berg and Smalla, 2009). However, these topological metrics, such as changes in node/edge numbers and connectivity, should be viewed as statistical indicators of community reorganization rather than direct proxies for ecological stability or functional capacity (Olesen et al., 2007). Our study links these network reorganizations to the post-topping environment and provides a community-level perspective on microbial dynamics following specific agricultural interventions. Further functional validation is required to elucidate the underlying mechanisms driving microbial dynamics.

Temporal changes following topping were accompanied by shifts in the phytohormone balance in tobacco roots, with significant increases in IAA and ABA and a decrease in GA8 (Fig. S3). These hormonal shifts were indicative of developmental transitions and were linked to plant resource allocation, defense mechanisms, and the biosynthesis of secondary metabolites like nicotine (De Boer et al., 2011; Hou et al., 2010; Lackman et al., 2011; Qin et al., 2020; Shoji et al., 2000). Previous studies have separately linked topping to nicotine accumulation (Zhang et al., 2023) and to phytohormone changes (Hou et al., 2010; Lackman et al., 2011; Shoji et al., 2000), but few studies have integrated these plant physiological responses with simultaneous temporal changes in root-associated microbial communities across multiple compartments. Our study bridges this gap by measuring phytohormones, alkaloids, and microbiome composition in parallel. Elevated IAA and ABA levels may be associated with stress-related pathways and may facilitate resource redirection towards alkaloid synthesis, including nicotine. Concurrently, nicotine and its precursor putrescine increased significantly, reaching their peak levels several days following topping (Fig. 4D) (Zhang et al., 2023). This temporal coupling between topping, hormonal changes, and nicotine biosynthesis provides a plausible mechanistic link for

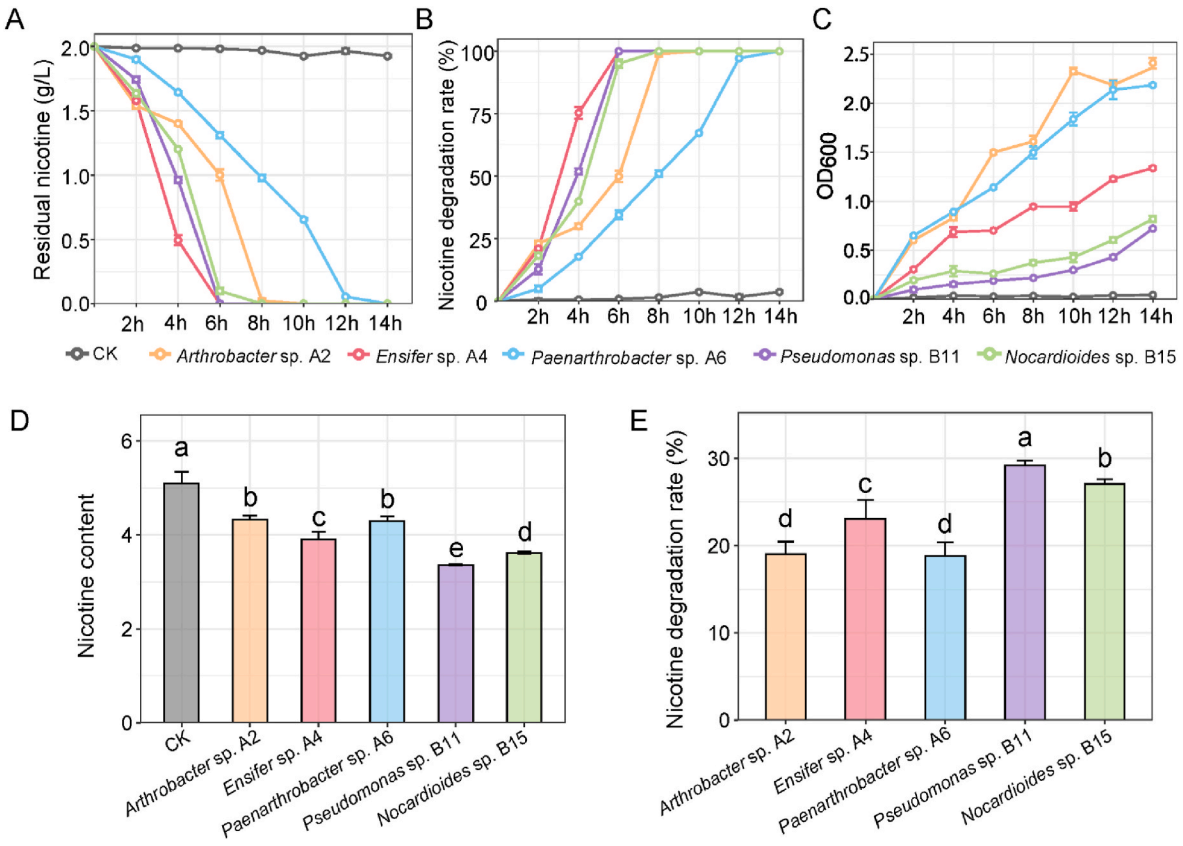


Fig. 8. Detection of nicotine degradation efficiency. (A) Residual nicotine content after treatment with nicotine-degrading bacteria. (B) The nicotine degradation rate of nicotine-degrading bacteria. (C) Growth curve of nicotine-degrading bacteria in nicotine medium. (D) The nicotine content in tobacco leaves after treatment with nicotine-degrading bacteria. (E) The nicotine degradation rates in tobacco leaves after treatment with nicotine-degrading bacteria.

understanding subsequent microbial responses. This substantial accumulation of nicotine in root tissues was accompanied by a notable enrichment of specific bacterial genera previously implicated in nicotine degradation: *Paenarthrobacter* in bulk soil, *Streptomyces*, *Bacillus* and *Arthrobacter* in the rhizosphere, *Ensifer* in the rhizoplane, and *Nocardioides* in the root endosphere (Fig. 3C) (Liu et al., 2015; Mu et al., 2020). In the root endosphere, *Nocardioides* (Ganas et al., 2008) was relatively more abundant than *Pseudomonas* (a well-known nicotine-degrading genus) (Xia et al., 2018). These patterns corroborate previous studies and suggest a novel association between nicotine-related compounds and the enrichment of taxa capable of nicotine metabolism, emphasizing the potential for targeted bioremediation strategies.

Correlation analysis provided associative evidence linking these metabolic and microbial changes. Specifically, nicotine, its precursor (putrescine), and its derivative (nicotine N-glucoside) exhibited significant correlations with various bacterial genera across ecological niches (Fig. 5). Notably, nicotine-degrading bacterial genera (*Paenarthrobacter*, *Arthrobacter*, *Streptomyces*, *Nocardioides*, and *Ensifer*) were positively correlated with nicotine or putrescine levels (Fig. 5), suggesting that elevated nicotine and related alkaloid levels may relate to the enrichment of taxa capable of nicotine metabolism (Berg and Smalla, 2009). The increased nicotine degradation products (e.g., 6-hydroxynicotine) in roots at later time points (Fig. 4D) further supported active catabolism by these microbes. While correlation analyses offered valuable associations, they did not establish causality. To further examine this relationship, we conducted a microcosm experiment by introducing exogenous nicotine into field soil. This experiment revealed that the decrease in extractable nicotine concentrations was accompanied by a significant increase in the relative abundance of key bacterial taxa (*Paenarthrobacter*, *Arthrobacter*, *Streptomyces*, *Ensifer*, and *Nocardioides*). This experimental evidence moves beyond correlative observations, indicating a potential association between nicotine availability and the enrichment of taxa known to possess nicotine-degrading capabilities. Compared with previous studies that isolated nicotine-degrading bacteria without explicitly linking them to post-topping plant metabolite dynamics (Liu et al., 2015; Mu et al., 2020), our work provides an ecological context by connecting field temporal correlations with a controlled nicotine-addition microcosm. However, given the complexity of soil processes, the specific roles of these microbes in nicotine transformation versus other abiotic factors remain to be fully disentangled. For instance, other components in the root exudates may also play crucial roles in microbial dynamics and function. Analyzing the post-topping temporal changes in root exudate composition could be a promising avenue for future research.

The enrichment of specific microorganisms under stress conditions may benefit plant survival and adaptation (Hernandez et al., 2021; Liu et al., 2020). For example, drought-stressed rice consistently enriched *Actinobacteria* and *Chloroflexi* while depleting *Acidobacteria* and *Deltaproteobacteria* (Santos-Medellin et al., 2021). Similarly, in our study, increased nicotine levels after topping were associated with the enrichment of nicotine-degrading microorganisms, which was supported by the isolation of nicotine-degrading bacteria *Arthrobacter* and *Nocardioides* from post-topping roots (Fig. 7B). Importantly, isolates from these enriched genera could efficiently degrade high concentrations of nicotine (Fig. 8B). Five selected strains (*Arthrobacter* sp. A2, *Ensifer* sp. A4, *Paenarthrobacter* sp. A6, *Pseudomonas* sp. B11, and *Nocardioides* sp. B15) demonstrated rapid nicotine degradation (2 g/L within 6–14 h), faster than previously reported strains such as *Pseudomonas putida* S16 (0.11 g/L/h) (Hu et al., 2019), highlighting their superior catabolic efficiency and expanding the candidate pool for nicotine bioremediation. Interestingly, slow-growing strains such as *Ensifer* sp.

A4, *Nocardioides* sp. B15, and *Pseudomonas* sp. B11 still displayed relatively fast nicotine degradation, indicating that growth rate and catabolic efficiency were not necessarily linked and may reflect different expression of nicotine-degrading enzymes (Brandsch, 2006)*Ensifer* sp. A4 and *Pseudomonas* sp. B11, with the highest *in vitro* degradation rates (Fig. 8B), also reduced nicotine content in tobacco leaves (Fig. 8D and E), pointing to highly efficient nicotine degradation pathways. These strains deserve further investigation as potential bioremediation agents in tobacco cultivation and processing.

Notably, *Ensifer* sp. A4 demonstrated marked potential for nicotine degradation in soil, reducing its concentration from 50 mg/kg to 1 mg/kg within 48 h post-inoculation, while the degradation rate was slower in uninoculated controls (Fig. S8). This rapid degradation aligned with the established role of soil microorganisms in pollutant breakdown and reinforced the notion that nicotine may be associated with the enrichment of specific microbial taxa (Anwar et al., 2009)However, these results were obtained from sterilized soil under controlled conditions. In complex field environments, factors such as microbial competition, nutrient limitations, and environmental variability are expected to substantially impede degradation efficiency. Therefore, while the microcosm data suggest promising potential for in-situ bioremediation, any extrapolation to field conditions remains tentative and requires rigorous field validation. Nevertheless, a simplified first-order estimate (assuming a conservative 90% reduction in efficacy under field conditions) would still predict a time frame on the order of weeks, but this is a coarse, back-of-the-envelope calculation that should not be interpreted as a definitive prediction. Our future studies will prioritize field experiments under natural conditions to evaluate the practical applicability of these strains in nicotine-contaminated environments.

We acknowledge that the absence of a non-topped control group limits our ability to definitively attribute all observed dynamics solely to topping. However, this limitation is substantially mitigated by the integrated multi-omics and experimental approach. Our study provides a well-resolved temporal association between nicotine accumulation and microbial community shifts, supported by functional evidence of nicotine-degrading capacity. This framework offers a robust foundation for developing targeted strategies to mitigate nicotine accumulation and promote soil health.

5. Conclusions

This study provides a novel temporal perspective on the dynamic interplay among tobacco topping, plant physiology, and root-associated microbiomes. It revealed that following topping, the structure and composition of microbial communities in the bulk soil, rhizosphere, and root endosphere underwent temporal changes. Furthermore, the temporal enrichment of highly efficient nicotine-degrading bacterial genera, supported by strain isolation and microcosm validation, suggested that host-derived metabolites may selectively recruit functionally specialized microbial taxa. These findings expand our understanding of plant-microbe-metabolite interactions in agricultural systems and offer valuable microbial candidates for mitigating nicotine accumulation.

CRediT authorship contribution statement

Mengli Gu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Jingjing Jin:** Writing – review & editing, Visualization, Data curation. **Huan Su:** Writing – review & editing, Visualization, Data curation. **Peng Lu:** Writing – review & editing, Visualization, Data curation. **Mengmeng Kong:** Writing – review & editing, Validation. **Zechao Qu:** Writing – review & editing, Validation. **Junjia Guo:** Writing – review & editing. **Sujie Wang:** Writing – review & editing. **Mengxiao Xu:** Writing – review & editing. **He Li:** Funding acquisition, Conceptualization. **Jianfeng Zhang:** Writing – review & editing. **Heli Shi:** Writing – review & editing. **Bikun Xiang:** Writing – review & editing. **Peijian Cao:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis. **Jiemeng Tao:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Consent for publication

Not applicable.

Availability of data and material

The sequencing data generated and/or analyzed during the current study are available in the Genome Sequence Archive in National Genomics Data Center, Beijing Institute of Genomics, Chinese Academy of Sciences (https://bigd.big.ac.cn/gsa), under accession numbers CRA024968, CRA037482 (16S), and CRA024970 (ITS). The metabolomics datasets are available in the OMIX, China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences (https://ngdc.cncb.ac.cn/omix/), under accession numbers OMIX010297 (Alkaloids) and OMIX010296 (Phytohormones).

Funding

This work was supported by the Chief Scientist Innovation Project of State Tobacco Monopoly Administration/China National Tobacco Corporation (912025CK0520, 912025CK05210), the Natural Science Foundation of HeNan Province (262300420470), Zhengzhou Tobacco Research Institute (902023CR0020, CNTC: 110202502023, 110202201001 (JY-01), 110202102050), Beijing Life Science Academy (BLSA: 2024200CC0130, 2023000CC0110), the Scientific and Technological Project of Henan Province (222102110224).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Not applicable.

List of abbreviations

BS	Bulk soil
RS	Rhizosphere
RP	Rhizoplane
RT	Root endosphere
ABA	Abscisic acid
GA	Gibberellin
IAA	Auxin
JA	Jasmonic acid
PCR	Polymerase chain reaction
PBS	Phosphate Buffered Saline
TSB	Tryptic Soy Broth
ASV	Amplicon Sequence Variant
PCA	Principal component analysis
N	Niche
S	Stage
MSM	Mineral salts medium
PERMANOVA	Permutational Multivariate Analysis of Variance
ANOVA	Analysis of Variance
PLS-DA	Partial Least Squares Discriminant Analysis

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2026.149786.

Data availability

Data will be made available on request.

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